Neural Caption Image Retrieval

Two page project write up | 03 October 2026

Problem statement

Given a natural language caption, retrieve and rank the matching image from a gallery. The task links sentence meaning to visual content, although captions and images begin in different feature spaces. The project compares a linear caption to image mapping with recurrent models that learn a shared space for both modalities.

Dataset

The experiments use MS COCO 2014 images and the Karpathy split: 113,287 training images, 5,000 validation images, and 5,000 test images. The authors' feature archive provides five captions per image and normalized 4,096 dimensional VGG 19 fc7 features obtained with ten crops. All feature rows were checked against the Karpathy caption order. Following the released source, training captions and the first 1,000 validation images define a vocabulary of 27,011 entries including two special tokens. Whitespace tokens are padded to 50 positions. Pretrained text features use 300 dimensional GloVe 6B vectors; missing vectors start at zero.

Approach

Linear baseline. Average available GloVe vectors in each caption, then fit a multiple response least squares map from 300 dimensional caption vectors to VGG image features. Rank images by squared Euclidean distance from the mapped query.

Neural models. Project VGG features into a 1,024 dimensional space. Encode caption words with either a GRU trained from scratch, a GRU initialized with GloVe, or an LSTM initialized with GloVe. The image projection includes a bias, and the recurrent final state represents the caption. Both embeddings are normalized to unit length. Similarity is the negative squared Euclidean distance between those representations. A bidirectional margin ranking loss (margin 0.1) separates matching image caption pairs from mismatched pairs; other captions of the same image are not treated as negatives.

Training. Each neural model completed 50 epochs with Adam, batch size 128, and an explicit learning rate of 0.001 and 886 full batches per epoch. Each cycle samples one of five captions per training image. The ranking loss sums incorrect pairs. The report lists 0.05, but the released code never passes that field to Adam. The final run follows legacy Adam updates and selects the best checkpoint using validation Recall@10 only. Test scores do not select model settings.

Retrieval path

For a new sentence, the system applies the training tokenizer and encodes its words with the selected caption model. It compares that representation with normalized projected VGG features from the saved test gallery, sorts by negative squared distance, and returns the top ranked image paths and scores. Saved image features are reused for each query. The three neural variants and the linear baseline use the same image gallery.

Implementation overview

The repository verifies and imports the original feature archive, prepares manifests and vocabulary, fits the baseline, trains recurrent variants, and evaluates retrieval. The GRU and LSTM follow legacy Keras gate equations and process all 50 padded positions. Feature arrays are mapped into memory; training saves progress and random state for resumption. The README provides setup, evaluation, and live retrieval commands.

Evaluation

Caption to image retrieval is evaluated on a fixed gallery of 1,000 test images, with one selected caption per image for each standard method (1,000 queries). Reported measures are Recall@1, Recall@10, and mean image rank. The pooled caption baseline combines the first five held out captions for each image into one query and is identified separately from single caption retrieval. Additional analysis checks recall by caption length, recall across training epochs, and short/long query examples.

Validation and example

Validation selected epochs 46, 46, and 38 for GRU, GRU + GloVe, and LSTM + GloVe. Each validation pass uses exactly 1,000 images and 5,000 caption queries. All three 50 epoch histories and every test gallery boundary were checked. For the query 'a guy riding a bike next to a train,' GRU + GloVe placed the cyclist beside a train at rank 1. The short query 'dog' returned a dog at rank one. Both rankings were reproduced from saved models.

Results

Method	R@1	R@10	Mean rank
Average GloVe baseline	9.5%	45.2%	37.06
Baseline + pooled captions*	11.0%	52.2%	25.64
GRU	34.5%	82.8%	9.13
GRU + GloVe	34.5%	82.6%	9.70
LSTM + GloVe	31.7%	83.1%	8.64

* Pooled caption baseline combines five test captions per image; other methods use one selected caption per image.

Conclusions

On the fixed 1,000 image gallery, the LSTM with GloVe achieved the highest single caption Recall@10 (83.1%). The highest Recall@1 was 34.5%. LSTM with GloVe had the lowest mean rank (8.64). Separately, the released evaluator's 5,000 caption protocol gives neural Recall@10 of 84.52%, 85.04%, 84.96%, in GRU, GRU + GloVe, and LSTM + GloVe order, versus 85.8%, 86.8%, and 86.2% in the paper. The final run follows legacy Adam and generator conventions. Original checkpoints, seeds, software versions, and baseline code are unavailable. Scores remain close for the neural models, but exact numerical reproduction has not been achieved.

Code: github.com/kishansaaai/ML_PROJECT. Source: Qian and Lamberti, *Neural Caption Image Retrieval* (CS229).